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The economics of short rotation coppice in Germany

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ABSTRACT

Short Rotation Coppice (SRC), which entails managing wood plantations as perennial energy crops on agricultural land, has the potential to contribute significant amounts of wooden biomass to Europe's energy mix. Yet, uncertainty prevails about the future of key economic variables determining the viability of SRC cultivations. Consequently, agriculturists face challenges when conducting ex-ante economic analyses of SRC projects and measuring performance against the alternative, agricultural crops. This paper scrutinises five key determinants of SRC economic viability: yield level, woodchip market price, subsidies, cost level and opportunity costs for conventional agricultural crops. By utilising site-specific conditions and different future scenarios we provide a comprehensive economic appraisal of SRC plantations for agriculturists. We focus our analysis on Germany. Our results show that SRC plantations are less profitable under the medium scenario when compared with agricultural crops. Notwithstanding, favourable political and economic conditions such as subsidies, lower costs and higher woodchip prices can lead to SRC's superior profitability. If the German government is serious about improving investments conditions for commercial SRC plantations, we recommend introducing sufficient, efficient and consistent subsidies.

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1. Introduction

Europe faces the threefold challenge of meeting increasing energy demands while reducing its dependency on fossil fuels and mitigating climate change. To effectively address these challenges, the European Union (EU) aims to support the development of renewable energy sources. This is embodied in Directive 2009/29/EC, which states that at least 20% of total energy consumption should be met by renewables by 2020. (10% in the transport sector) [1].

This target still requires ample efforts from Member States. In 2009, only 9% of EU's gross inland energy consumption was from renewable sources, of which biomass contributed the largest share (68.6%) [2]. Given its high-energy content and versatile use for electricity and heat generation, wood is

a preferred source of biomass. Most wooden biomass is sourced from conventional forestry or directly derived from existing wood residues or industrial by-products.

An alternative option for wood biomass sourcing is Short Rotation Coppice (SRC). This entails managing wood plantations as perennial energy crops. Common plants suitable for SRC include poplar (*populus spp.*), willow (*salix spp.*) and miscanthus (*miscanthus spp.*). These are planted on agricultural land and harvested in 2–5-year rotation cycles. The total operation length varies between 20 and 30 years concluding with a re-cultivation of the planted land [3–5]. SRC offers a number of advantages relative to competing agricultural crops: it requires less operational efforts; provides soil protection against wind and water erosion; limits nutrient leaching and requires reduced fertilisation. As a result, it has

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been argued that SRC improves the sustainability of agricultural production [6,7].

Despite these advantages, plantations managed under SRC are still marginal in Europe. Sweden is the largest SRC operator (16,000 ha) followed by Poland (9000 ha), UK (6000 ha), Italy (5000 ha), Germany (5000 ha) and France (3000 ha) [7]. The lack of commercially driven SRC establishment is partially due to operators seldom achieving economically attractive yield levels under current woodchip prices [8,9].

Furthermore, much uncertainty prevails about the future of key economic variables determining the viability of SRC cultivations, such as woodchip prices or subsidies. Consequently, agriculturists face challenges when conducting ex-ante economic analyses of SRC projects and measuring performance against the alternative, annual agricultural crops, with an investment horizon of one year or less [10,11]. An additional merit of agricultural crop plantations in contrast to SRC is that agriculturists can choose what crops to grow from year to year based on their current market prices. SRC sites as long-term investments limit the opportunity to benefit from attractive agricultural crop prices.

Despite increased interest in SRC as a potentially viable economic activity, the existing literature falls short of providing reliable economic benchmarks for individual SRC agriculturists. Below, we review existing literature along the five key determinants of SRC economic viability: yield level, woodchip market price, subsidies, cost level and opportunity costs. This paper aims to enhance the assessment of these key variables in order to facilitate a comprehensive economic appraisal of the competitiveness of SRC. The applied methodology considers site-specific conditions and conducts scenario analysis.

We focus our SRC analysis on Germany. The reason for this is that future political and economic developments are likely to turn SRC operations more attractive: First, a governmental scenarios analysis predicts that 450,000 ha could be planted with SRC by 2020 [12]. In order to achieve this, subsidies as well as governmental funding to private research initiatives for SRC might be necessary. In addition, increasing demand for wooden biomass can be expected [7]. Finally, large energy suppliers have recently started to develop commercial SRC projects as a means to supply their combined heat and power plants.

In the following, the five key economic variables for SRC operations will be introduced.

1.1. Yield level

The yield level significantly affects revenues of SRC plantations. Yield estimation depends on both managerial decisions and site-specific conditions. To avoid destructive sampling, empirical non-destructive SRC yield models are commonly used [13,14]. Model design and application are however often country-specific. For instance, Mola-Yudego & Aronsson's model [8] is based on a district-specific agro-climatic index in Sweden and Evans et al.'s model [15] requires the provision of UK grid references. In Germany, Wael [16] and Murach et al. [17] developed SRC yield estimators, yet these suffer from a number of limitations for the applications considered in our study. While Wael's model is adapted for different poplar genotypes, varying rotation lengths and site-specific data (soil

quality, temperature and water supply), it only generates yield forecasts for the first harvest. The effect of increasing biomass production levels in subsequent rotations as a result of increased SRC rooting systems is neglected [5,18,19]. The same shortcoming applies to Murach et al.'s model. In addition, Murach et al.'s model exclusively forecasts yield depending on available water supply while assuming site conditions and genotype choice to be optimal.

In response to this, we propose a new yield estimator that predicts long-term yield levels for three poplar and one willow species in 3-year rotation cycles. Differing site conditions and planting densities are integrated as parameters to the estimator.¹

1.2. Woodchips market price

The second key determinant impacting SRC revenue is woodchips market price. Between 2003 and 2010, real woodchip prices in Germany have increased by 56% (from 14.9 €/MWh to 23.2 €/MWh). This is a result of soaring demand and rising heating oil prices, advances in bioenergy technologies and political support for renewables. Price growth is expected to continue, albeit at lower pace. This would imply higher revenues for a given amount of produced biomass. Agriculturists are moreover interested in price developments of agricultural crops in relation to woodchip. The OECD and FAO for instance expect cereal prices to decline until 2020 which would turn SRC sites economically more attractive [20]. There is high uncertainty about both future woodchip and agricultural crop prices: key forces driving price volatility entail the evolution of the oil price, climate change legislation, improvements in yield levels and the adaptation of agriculture to climate change.

Despite the importance of woodchip and agricultural crop price volatility on SRC profitability, this impact has not been fully considered in the literature. In this study, woodchip prices were forecasted based on different heating oil price scenarios derived from the Energy Information Administration (EIA) [21].

1.3. Subsidies

The third determinant considered is subsidies for SRC plantations. For instance, the Federal State of Saxony reimburses up to 30% of initial SRC investments (expenditure related to site preparation, cuttings plantation and fencing) [11]. The Energy Crop Scheme in England provides a similar subsidy, which covers 50% of all eligible SRC expenses [22]. Subsidies can also be granted in the form of fixed area payments, as in Sweden [23]. In this case, operators are given a payment per hectare of planted SRC. Both subsidy designs and their impact on SRC profitability are explored in this paper.

1.4. Cost levels

Cost related estimates to SRC operations vary widely and are not comprehensively reviewed in the literature. This creates

¹ It should be noted that the estimator can also be applied in countries outside Germany. This will be shown in 2.1.

Table 1 – Overview pilot projects (P=Precipitation, T = Temperature, SQI = Site quality index, N=Number).

Source	Location	P mm	T °C	SQI	Plant density (N/ha)	Selected poplar clones	Selected willow clones
Werner et al. [19]	Dornburg	578	7.5	60	11,000	Unal, Raspalje, Beaupré, Donk, Androscoggin, Max	722/51
	Langen wetzendorf	650	8.8	42	11,000	Unal, Boelare, Beaupré, Donk, Androscoggin, Max	722/51
Boelcke [5]	Bad Salzungen	586	8.1	32	6900	Androscoggin, Max, NE 42	Tora
	Gülzow	532	8.2	48	13,330	Rap, Max, 10/85	Zieverich, Königshanfweide, 78-021, 78-101, 78-183, 57/57, Björn, Rapp, Ulv Weide 10
	Vipperow	640	8.0	30	13,330	Max	
Grunert [24]	Krummenhennersdorf	820	7.2	45	11,850	Max, Hybrid 275	Jorr
Maier & Vetter [25]	Mühlheim	650	9.5	81	10,000	Only species is defined	Only species is defined

risks of costs over or underestimation. In addition, cost levels for SRC sites might alter in the future: an increasing number of commercial plantations are likely to result in decreasing planting and harvesting costs due to learning effects while higher energy prices are likely to lift fertiliser and transportation costs.

Hence, to provide a thorough picture, cost data was retrieved from 14 recent sources to determine full costs. The median, minimum and maximum values for all relevant cost pools were derived from this compilation. Based on these, we provide different cost scenarios and an assessment of their impact on SRC profitability.

1.5. Opportunity costs

Finally, the economic performance of SRC must be measured against the main alternative investment: agricultural crops [10]. A benchmark annuity (*annual gross margin*) for agricultural crops was therefore created for poor, medium and good site conditions. This provides agriculturists with an approximate estimate of opportunity costs.

2. Methodology

In the next sections, we present the methodology for assessing the impact of the key determinants introduced above on the economic viability of SRC. It should be noted that the yield estimator is based on secondary field data from pilot projects managed by SRC experts. The estimations should therefore be considered as yield levels to be achieved under optimal managerial decisions.

2.1. Yield estimator

The estimator (a *non-destructive statistical yield model*) forecasts yield based on four key variable categories: species [poplar and willow]; rotation sequence [3-year rotation cycles]; site characteristics [average annual temperature (T), average

annual precipitation (P), average water capacity (WC), capillary moisture rise (CMR) and soil quality index (SQI)]²; and plant densities (5000 to 20,000 cuttings per hectare). All data are derived from SRC pilot projects in Germany.

2.1.1. Data collection

Yield data were collected from 7 pilot sites derived from 4 studies. All sites and their respective conditions, planting densities and planted clones are summarised in Table 1. P varies between 532 mm in Gülzow and 820 mm in Krummenhennersdorf. Soil textures were predominantly sandy loam explaining SQI values below 50. Solely Dornburg and Mühlheim have better soil qualities with SQIs of 60 and 81 respectively. The lowest and highest Ts are found in Krummenhennersdorf (7.2 °C) and Mühlheim (9.5 °C). Planting densities ranges from 6900 plants per hectare in Bad Salzungen to 13,300 plants per hectare in Gülzow and Vipperow. All studies provide yield data for at least 2 consecutive harvests in 3-years rotation cycles.

2.1.2. Categorising and constraining yields

The yield data was categorised along key yield determinants as follows: Firstly, for all pilot sites the yield rates were categorised based on species composition and rotation sequence. The considered species were *P. trichocarpa* x *P. deltoids*, *P. maximowiczii* x *P. nigra*, *P. maximowiczii* x *P. trichocarpa* and *S. viminalis*. As yield data were only available for up to 4 rotation sequences (12 operation years in 3-years rotation cycles), we assumed an average yield decrease by a factor 0.91 from the 5th to the 7th rotation³ inclusively. From the 8th rotation, yield further decreased by a factor 0.74. These factors were derived from Kröber et al. [11].

Secondly, yields were further characterised with site-specific biological and geographical conditions. A Site Suitability Index (SSI) was developed. SSI considers temperature, soil and rainfall water supply as well as soil quality (3). Measures and units are as follows: average annual temperature (T in °C); average water capacity (WC in mm) and capillary rising moisture rate (CRM in mm) are used to determine the average water amount provided by the soil (W in mm); average annual precipitation (P in mm) and Soil Quality Index (SQI) for the soil quality. T, W, P and SQI were each given

² In Germany, the unified index “Ackerzahl” (Soil Quality Index) signifies the average soil quality of distinct arable lands. Determining the SQI is conducted by compounding soil texture, productivity, geological origins, water supply and climatic conditions.

³ 5th rotation being from year 12–15.

a weight of 25%. Total water supply therefore explains 50% of SSI, a value consistent with Murach et al. [17] and Wael [16]. For each variable (yield determinants), both a minimum and maximum values were defined. If site values transgress these boundaries, the set minimum or maximum value is assigned. The computation of SSI ($0 < \text{SSI} < 10$) for a rotation period t is shown in Formula 1.

$$\text{SSI}_t = 0.25 * 10 \left(\frac{T_t - \text{Min } T_t}{\text{Max } T_t - \text{Min } T_t} + \frac{W_t - \text{Min } W_t}{\text{Max } W_t - \text{Min } W_t} + \frac{P_t - \text{Min } P_t}{\text{Max } P_t - \text{Min } P_t} + \frac{\text{SQI}_t - \text{Min } \text{SQI}_t}{\text{Max } \text{SQI}_t - \text{Min } \text{SQI}_t} \right) \quad (\text{Form. 1})$$

Based on Landgraf, Johne & Röhle [26], DEFRA [3] and Wael [16], the minimum and maximum values for the respective yield determinants are as follows: $5^\circ\text{C} \leq T \leq 10^\circ\text{C}$; $60\text{mm} \leq W \leq 300\text{mm}$; $400\text{mm} \leq P \leq 1000\text{mm}$; $20 \leq \text{SQI} \leq 60$.

A site with a T of 7.5°C , a W of 180mm , a P of 700mm and a SQI of 40 would be classified as average with a SSI of 5 . A linear relation between yield and SSI is assumed [17]. An increase in SSI from 5 to 10 for a given species and same planting density would thus double the SRC yield.

Thirdly, consistent with Murach et al. [17] and Hartmann [14], it was assumed that yield increases linearly by factor 1.5 when planting density doubles. Minimum and maximum planting densities were set to 5000 and $20,000$ plants per hectare respectively. A crop shortfall rate of 6% was also assumed [27].⁴

2.1.3. Statistical testing

After categorising and constraining the yields, statistical testing was conducted to identify significant outliers which might be the result of flaws in biomass estimation or irregular biological/geographical conditions in a particular rotation sequence. A cross-sectional parametric t -test was applied under the assumption that the adjusted yields⁵ (AdY) were normally distributed.

- The null hypothesis H_0 is: The adjusted yield does not differ significantly from the species' average
- The alternative hypothesis H_A is: The adjusted yield differs significantly from the species' average

The applied statistics for a distinct species in rotation period t is:

$$t_{\text{AdY}} = \sqrt{n} * \frac{(\text{AdY}_t - \overline{\text{AdY}})}{S(\text{AdY})} \quad (\text{Form. 2})$$

$$\text{where } S(\text{AdY}) = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (\text{AdY}_i - \overline{\text{AdY}})^2} \quad \text{and} \quad \overline{\text{AdY}} = \frac{1}{n} \sum_{i=1}^n \text{AdY}_i$$

AdY with t -values above the 99% -level of significance were eliminated from the sample.

⁴ It can be noted that the SSI can be used for countries outside Germany by fixing SQI levels depending on respective soil textures.

⁵ Adjusted yield is derived from categorising and constraining yields (2.1.2).

2.1.4. Exemplary yield estimations

We tested the approach by forecasting hypothetical yields for a 24 year SRC site with 3-year rotation cycles. Poplar and willow species were planted at a $12,000$ and $17,500$ plants per hectare density, respectively.⁶ The results were then validated with findings from the literature.

Fig. 1a shows the yield levels for different species with respect to the SSI. Sites of SSI 5 yield between $5.7 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$ for *P. trichocarpa* x *P. deltoides* and $11.5 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$ for *P. maximowiszi* x *P. nigra*. The species *P. maximowiszi* x *P. trichocarpa* also provides high-yield rates of $10.3 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$, while *S. Viminalis* only achieve to reach levels of $7.8 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$. Under optimal sites conditions (SSI 10), yields amount to $23.1 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$ for the strongest species *P. maximowiszi* x *P. nigra*. The weakest species, *P. trichocarpa* x *P. deltoides*, achieves a yield level of $11.5 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$. On the poorest sites (SSI 1), *P. maximowiszi* x *P. nigra* plants reach a yield level of $2.3 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$. These forecasts are consistent with those found in the literature. Scholz et al. [28], estimated average yield levels for poplar and willow under medium site condition. His results ranged between 8 and $12 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$ and 5 – $9 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$ respectively. The poplar species *P. nigra* x *P. maximowiczii* and *P. maximowiczii* x *P. trichocarpa* are also found to yield the highest productivity [29].

Fig. 1b illustrates yield levels throughout 8 rotation sequences for different species when the SSI amounts to 5 . Due to stronger rooting systems, the average yield more than doubles between the 1st and the 4th rotation sequences (4.6 – $10.7 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$, respectively). For the remaining operation length, yield growth factors were derived from Kröber et al. (11) (see section 2.1.2). Yield growth rates reach an average 2.5 after the first harvest for *P. maximowiszi* x *P. nigra* while lower yield rates are obtained for *P. trichocarpa* x *P. deltoides* and *P. maximowiszi* x *P. trichocarpa* (2.3 and 2.1 respectively). *S. Viminalis* in contrast, has the lowest rates of all, averaging 1.6 after the first rotation.

Yields as a function of planting densities and for SSI 5 are shown in Fig. 1c. To achieve a yield level above $10 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$, poplar sites must be planted at a density above $10,000$ plants per hectare. Willow species requires an even greater planting density to achieve similar yield rates as poplar. Even at densities of $20,000$ plants per hectare, *S. Viminalis* plants only yield $8.6 \text{ t}_{\text{atro}} \text{ ha}^{-1} \text{ a}^{-1}$.

2.2. Woodchips market price forecast

Two main approaches are eligible to forecast woodchips market price: time series analysis (e.g. moving average, exponential smoothing) or econometric forecasting (regression analysis). The first one relies on historic pricing which is a poor basis for woodchip price forecasts. First, woodchip prices need to be forecasted for the entire operation length of SRC projects, that is to say 20 – 30 years. However, historic woodchip price data is only available for the period 2003 to 2010 which is insufficient to guarantee a reliable long-term forecast. Secondly, real price levels in 2010 were 58% higher

⁶ Due to higher accretion rates and lower average yields, willows require higher planting densities.

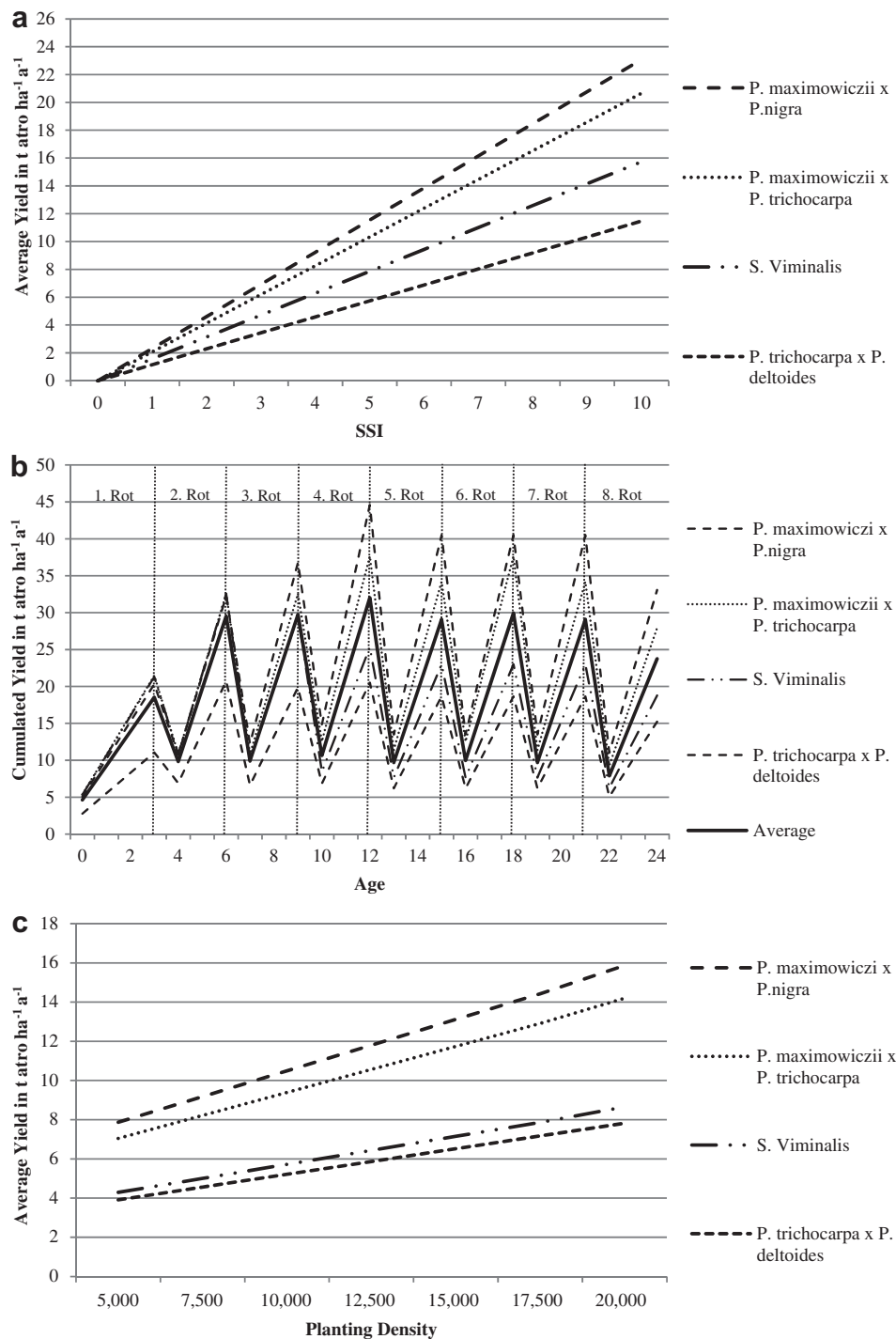


Fig. 1 – a–c Results of the yield estimator for *P. maximowiczii* x *P. nigra*, *P. maximowiczii* x *P. trichocarpa*, *P. trichocarpa* x *P. deltoides* and *S. viminalis* depending on site conditions (Fig. 1a); rotation sequence at medium site suitability (SSI = 5) (Fig. 1b) and planting density at medium site suitability (SSI = 5) (Fig. 1c).

than in 2003. Such price increases are unlikely to be representative of future prices, as one can assume that technologies and markets will mature.

In contrast, econometric forecasting enables the consideration of surrogate variables determining pricing. In this study, the pricing of substitute products is used to predict woodchip price. If the explanatory power is statistically

significant, the forecast can be based on the independent variable through regression analysis.

The following variables were tested for co-integration with woodchip prices to determine the level of explanatory power: (a) Heavy heating oil price for industrial at maximum sulphur rates of 1% for deliveries in excess of 15 tons in Germany [30]; (b) Light heating oil price for deliveries in excess of 500 tons

(wholesale) in Germany [30]; and (c) European Emission Allowance prices (in € per tonne CO₂) traded on the European Energy Exchange [31].

Only heavy heating oil prices showed good explanatory power at 95%-level of significance with a $r^2 = 0.97$. Woodchip prices were predicted using ordinary least square method (OLS) (2.2.2).

2.2.1. Co-integration testing and regression line

To test for co-integration, a simple Engle-Granger 2-step test was applied. By this means, it is tested if the residuals are stationary. If they are, it follows that woodchip prices per t_{atro} (W_t) [32] and heating oil prices per ton (Z_t) are co-integrated [33,34].

- The null hypothesis H_0 is: W_t and Z_t are not co-integrated
- The alternative hypothesis H_A is: W_t and Z_t are co-integrated

The null hypothesis can be rejected at the 95% significance level. The test statistic is -2.59 , i.e. lower than the critical value of -1.96 . It can therefore be concluded that there exists a long-run relationship in the historic price evolution of W_t

and Z_t [35]. By running regression of W_t on Z_t , the following regression line could be derived.

$$W_t = \alpha + \beta * Z_t = 45.6 + 0.15 * Z_t \quad (\text{Form. 3})$$

The projection values will be derived from the oil price forecast published in the Annual Energy Outlook 2011 by the U.S. Energy Information Administration (EIA) [21].

2.2.2. Results of the woodchip price forecast

Fig. 2 shows the forecasts for heavy heating oil and woodchip prices. Up to 2035, the real woodchip price is projected to grow by 26.7% and 78.1% under the reference and high oil price scenario respectively and decrease by 26.4% under the low oil price scenario. This translates in final prices of 29 € per MWh, 40.7 € per MWh and 17.2 € per MWh. The growth and decline rates for woodchips are weaker compared to the heating oil prices due to the relatively low level of sensitivity reflected in β in Formula 3. This result is reasonable: In the past, woodchips have not been a complete substitute for fossil fuels due to the secondary use of wood for material applications.

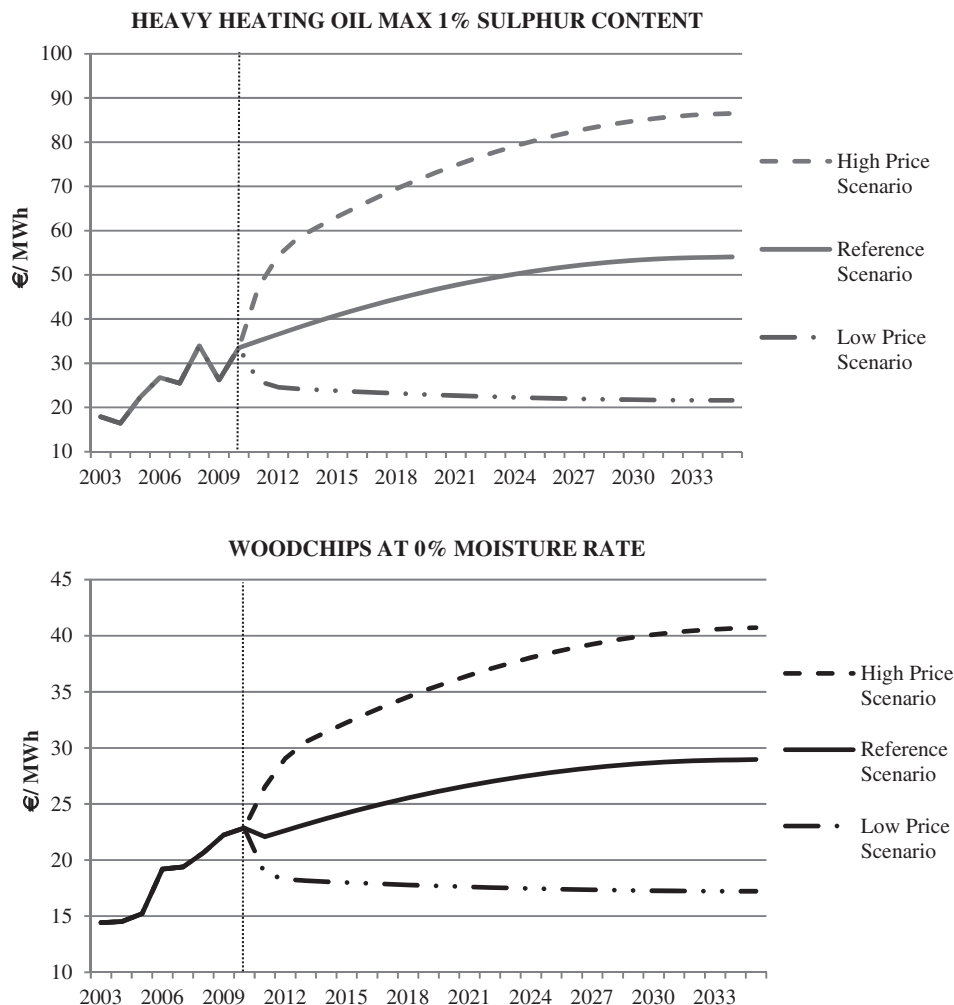


Fig. 2 – Heavy heating oil and woodchip price forecast for 2011 to 2035 (in 2010 € real) (21,32).

Table 2 – Overview of median, minimum and maximum cost estimates. Note that there was not sufficient data available to determine the min and max costs for “poplar own plantation” and “willow own plantation” source [5,11,19,24,27,36–44]

Cost estimates					
Cost Pool		Min	Median	Max	Unit
Preparation Costs	Ploughing	47,0	85,0	137,0	€/ha
	Soil Preparation		27,0	50,0	€/ha
	Weedcontrol (Flexidor 0,3 l + Fusilade 1 l)	50,3	70,3	72,9	€/ha
	Roundup 3l				
Plantation Costs	Flexidor 0,3 l	51,1	62,4	73,7	€/ha
	Fusilade 1 l	25,9	37,1	38,2	€/ha
	Cuttings	0,15	0,20	0,27	€/piece
	–Poplar Market		0,15		€/piece
	–Poplar Own Plantation				
	–Willow Market	0,06	0,08	0,11	€/piece
	–Willow Own Plantation		0,04		€/piece
Harvest Costs	Machinery Cuttings		0,04	0,09	€/piece
	Plantation				
	Harvesting & Logistics (incl. Transport)		20,4	25,0	€/t _{atro}
	Drying	15,0	15,1	15,1	€/t _{atro}
	Storage	5,0	5,3	6,1	€/t _{atro}
Re-cultivation	Fertilisers after Harvest		155,5	273,0	€/ha/rotation
		750,0	1023,0	1920,0	€/ha
	Additional Costs	105,0	170,0	250,0	€/ha
Additional Costs	Lease	100,0	139,5	154,0	€/ha
	Overheads	724	740	756	€/ha
	Fencing				

2.3. Cost estimator

The cost calculator is based on *full cost calculation* and retrieves data from 14 literature sources. For each cost pool and cost type the median,⁷ minimum and maximum values have been determined in order to investigate the economic impact of different cost levels on SRC plantations. For the “medium” cost scenario, the median values are taken and so on and so forth. Table 2 illustrates the relevant cost pools and their estimates expressed in their appropriate unit. Preparation, plantation and re-cultivation costs solely occur once while harvest costs obviously occur with every rotation. Running costs arise annually throughout the whole operation life cycle.

2.4. Economic appraisal

The decision to operate a SRC site depends on its economic prospects (return, net benefits and/or amortisation length). These dimensions are measured against the economic performance of annual crop cultivations. Both cultivation forms require the application of different evaluation techniques due to their differences in cash flow structures and project lifespan. Unlike agricultural crops, SRC are characterised by high initial costs, long-term irregular revenues and costs and operation lengths of 20–30 years. In this section, we therefore describe both the evaluation technique for agricultural crops and that for SRC. The decision criteria are the annuities for both crop types (for SRC, the annuity is referred to as NPV annuity and for agricultural crops, *benchmark annuity*). All monetary values are expressed in 2010 € per hectare.

⁷ The median was chosen instead of the mean value in order to minimise deviation bias.

2.4.1. Agricultural crops

Conventional agricultural crops are annual and have the same revenue and cost structures from year to year. Hence a *static appraisal* is used. Total annual revenues are deducted from total annual costs. Table 3 shows average yield, price and costs given specific site conditions for winter wheat, winter barley, winter rye and winter oilseed rape for 2011. These are crops most likely to be substituted by SRC [11,24,27]. By applying *static appraisal*, it is assumed that the benchmark annuity amounts to 230 € for sites with a SSI between 1 and 3.9; 460 € for sites with a SSI between 4 and 6; 710 € on sites with a SSI between 6.1 and 10.⁸

2.4.2. Dynamic appraisal for SRC sites

For SRC, cash-flows (CF) occur in the future thus need to be discounted (37). A *dynamic appraisal* (Formula 4) is therefore used. The denominator represents the discount factor for year *t* with the discount rate *i* (operation length of the SRC site from *t* = 0 to *t* = *n*):

$$\text{Net Present Value (NPV)} = \text{Initial Investment Costs}_{t=0} + \sum_{t=1}^n \frac{\text{Future Revenues} - \text{Future Costs}}{(1+i)^n} \quad (\text{Form. 4})$$

Through multiplication by the annuity factor, the Net Present Value (NPV) is transformed into a constant annual profit/loss value.

$$\text{Net Present Value Annuity}_{t=0 \text{ to } n} = \text{NPV} * \frac{(n+i)^n * i}{(n+i)^n - 1} \quad (\text{Form. 5})$$

⁸ Benchmark values are estimates only: yields, prices and costs vary not only throughout time but also between regions.

Table 3 – Overview yields, prices and costs for annual crops, source [45]:

Benchmark annuity for market Fruit cultivation depending on site conditions

	Site Conditions	Winter Wheat	Winter Barley	Winter Rye	Winter Oilseed Rape	Average
Yield ($t_{atro} \text{ ha}^{-1} \text{ a}^{-1}$)	Low	6	6	6	3	4
	Medium	8	8	7,5	4	7
	High	10	10	9	5	10
Price ($\text{€}/t_{atro}$)		210	175	180	395	240
Variable Costs ($\text{€}/\text{ha}$)	Low	650	610	640	700	650
	Medium	725	690	690	750	714
	High	800	720	740	790	763
Fixed Costs ($\text{€}/\text{ha}$)		327	327	327	327	327
Annuity ($\text{€ ha}^{-1} \text{ a}^{-1}$)	Low	368	184	114	238	226
	Medium	629	384	334	504	462
	High	890	634	554	780	714

A NPV annuity exceeding the annuity of agricultural crops implies that the decision maker should opt for cultivating SRC and vice versa.

3. Results

First we present the economic performance of the strongest SRC species under different site conditions (3.1). We then discuss how different subsidy (3.2), cost (3.3) and woodchip price scenarios (3.4) affect the NPV annuity. Our assumptions are as follows

- Species *P.maximowiszi* x *P.nigra* are planted at a density of 12,000 cuttings per hectare
- Total operation length is 24 years with 3-year rotation cycles
- A discount rate of 6% is assumed (10,11)

3.1. The economic performance of SRC under different site conditions

The site conditions were categorised as poor, medium and good (SSIs of 3, 5 and 7 respectively). Costs and woodchip price levels were assumed to be medium. No subsidies for SRC were considered.

Fig. 3a–c show the annual discounted cash flows (DCF); the cumulated DCFs; the NPV annuity and finally the benchmark crop annuity depending on site conditions. Under medium site conditions, the SRC site generates a NPV of 3830 € per hectare (see Fig. 3a). The NPV annuity amounts to 310 € per hectare and amortisation takes place in year 12. When assessed against the benchmark, the SRC site underperforms by 150 €. On poor sites, the SRC operator would incur a negative NPV of 990 € per hectare (see Fig. 3b). The SRC NPV annuity is clearly lower than the benchmark (difference of 310 €). On good sites, NPV reaches 8660 € per hectare. Amortisation occurs in year 9 and the annuity difference decreases to 20 €.

To conclude, SRC sites planted with the high-yield species *P.maximowiszi* x *P.nigra* deliver net benefits when site conditions are at least medium. Furthermore, SRC becomes increasingly profitable with improved site conditions comparatively to

agricultural crops. This is due to differences in cost structures whereas SRC plantations have a lower proportion of variable costs.

3.2. Woodchip price scenarios

The results presented in section 3.1 show that SRC NPV annuities do not reach the benchmark values. In this section, we assess the impact of woodchip price based on forecasts by the EIA (21) on this outcome. Under the EIA's medium oil price scenario (21), real woodchip prices are predicted to increase by 60% from 2010 to 2035. In contrast, the low oil price scenario assumes higher supply and lower demand levels relative to the medium scenario resulting in decrease in prices by 36% for the same period. The reverse applies to the high oil price scenario which assumes a price increase of 158%. Moreover, agricultural crop prices are also increasingly linked to the oil price in the long run. The OECD and FAO estimate that a 25% price increase of crude oil prices results in a 5% price increase for coarse grains, a 4% price increase for wheat and a 3% price increase for oilseed [20]. This would also improve the economic competitiveness of agricultural crops and is reflected in the high oil price benchmark values. Further, we assume medium cost levels and the absence of subsidies.

Fig. 4a displays the respective annuities depending on yield and woodchip price scenario. In the case of low a woodchip price development, the NPV annuity is only slightly positive on very good sites. A high woodchip price development implies that SRC cultivations outperform agricultural crops (even at higher prices) when the yield level transgresses $8 t_{atro} \text{ ha}^{-1} \text{ a}^{-1}$. Moreover, the annuity is entirely positive irrespective of the site conditions with values ranging from 120 € at $6 t_{atro} \text{ ha}^{-1} \text{ a}^{-1}$ to 1400 € at $16 t_{atro} \text{ ha}^{-1} \text{ a}^{-1}$.

3.3. Subsidy scenarios

Three different scenarios are discussed. Cost and woodchip prices levels are both assumed medium. The “no subsidy” scenario assumes no available SRC grants. Under the “low subsidy” scenario, an initial investment subsidy of 30% is assumed. For the “high subsidy” scenario, the 30% subsidy as well as an area payment of 200 € per hectare per annum are

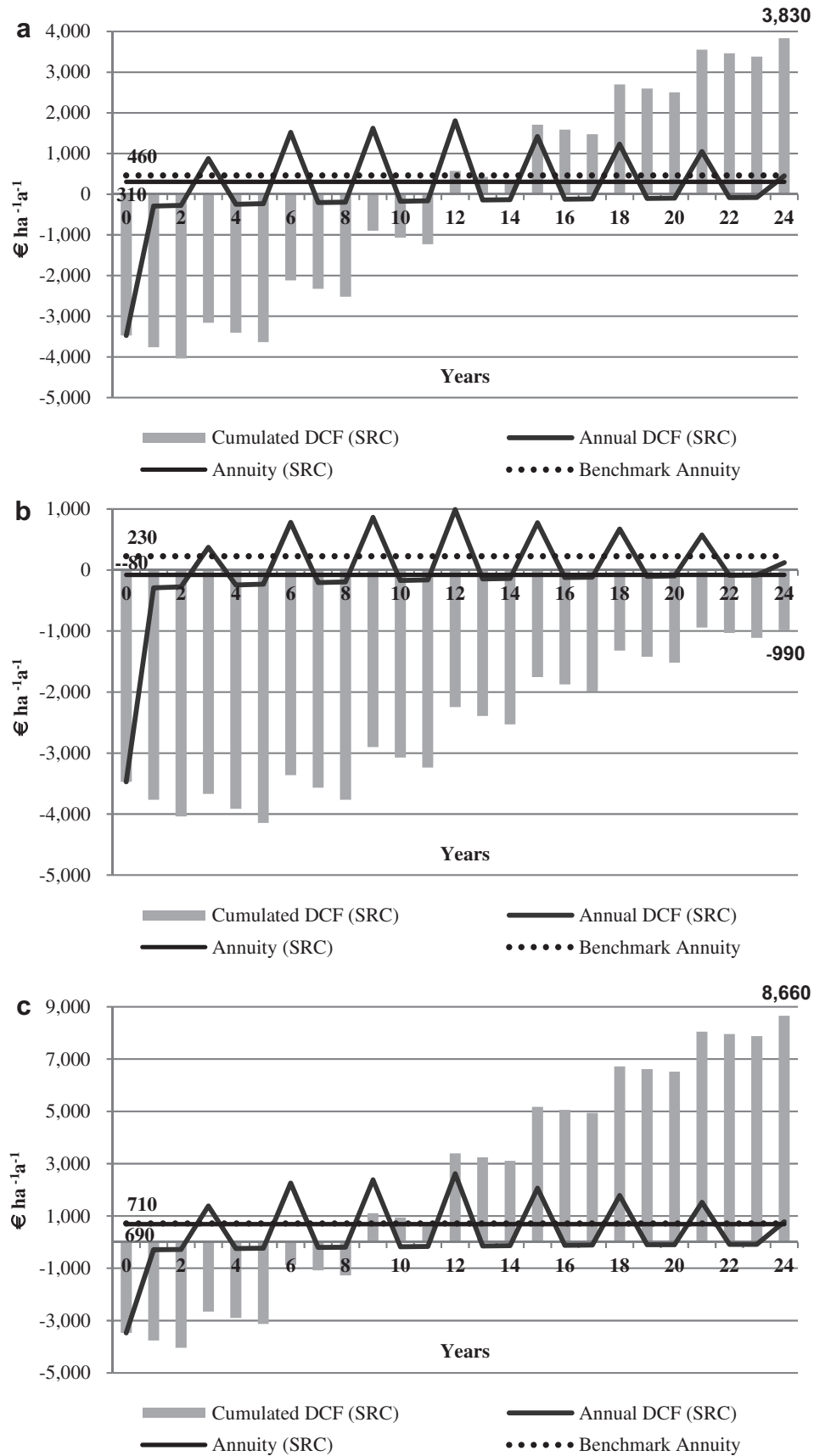


Fig. 3 – a–c: Discounted SRC cash flows and benchmark annuity for medium site conditions (SSI = 5) (Fig. 3a); poor site conditions (SSI = 3) (Fig. 3b) and good site conditions (SSI = 7) (Fig. 3c).

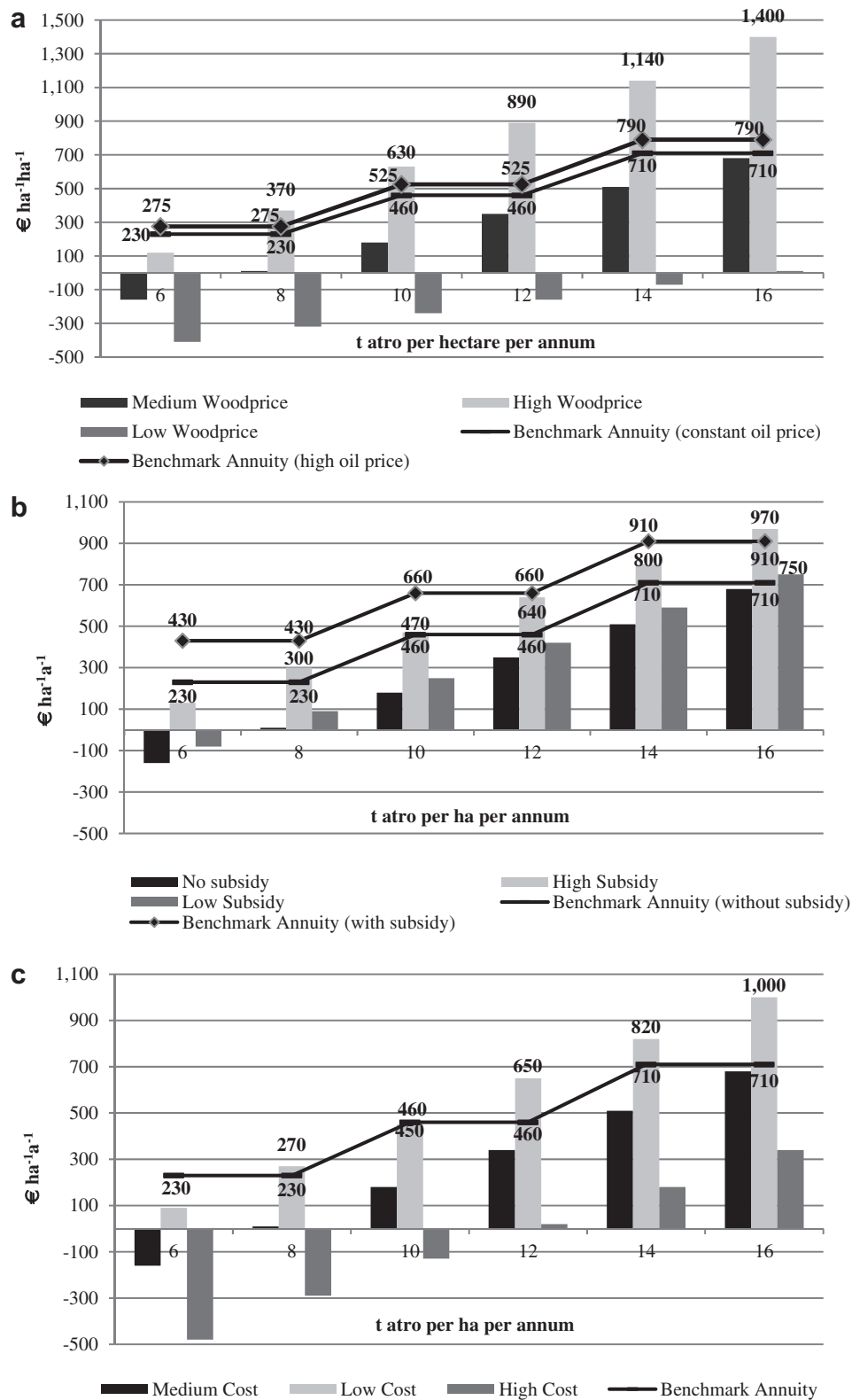


Fig. 4 – a–c: SRC annuity depending on woodchip price/yield level (Fig. 4a); subsidies/yield level (Fig. 4b) and cost level/yield level (Fig. 4c).

assumed. We further illustrate the benchmark annuity with an area payment of 200 € per hectare.

It should be noted that CO₂ allowances, as a market-based instrument, are an economically efficient alternative subsidy

to area payments. This is because allowances provide incentives to produce higher levels of biomass, as each saved ton of CO₂ is remunerated. A supposed CO₂ allowance price of 14 € per ton and a SRC yield level of 10 t_{atro} ha⁻¹ a⁻¹ (which saves

Table 4 – Comparison of the economics of SRC in Germany, Poland and Northern Ireland. Lease and overheads are excluded.

Assumptions & Results		Germany	Poland [10]	Northern Ireland [50]
NPV Annuity (Annual Gross Margin)	€ ha ⁻¹ a ⁻¹	570	259	386
Production Costs	€/t _{atro}	58	38	51
Woodchip price	€/t _{atro}	130	71	96
Lifespan	Years	21	22	22
Yield	t _{atro} ha ⁻¹ a ⁻¹	11.6	10	12
Discount Rate	%	6%	6%	6%

14.7 CO₂ tons per hectare [46]) would equal an area payment of roughly 200 € per hectare.

Fig. 4b shows the impact of the distinct scenarios on the SRC annuity for different yield levels. Under the low subsidies scenario, the NPV annuity remains below the benchmark for the considered yield range except at yield level 16 t_{atro} ha⁻¹ a⁻¹. This result changes when high subsidies are assumed. At a yield level of 10 t_{atro} ha⁻¹ a⁻¹, the benchmark value without subsidy can be surpassed by 10 € (benchmark annuity without subsidy). Above this level, SRC sites are always to be preferred over annual cultivations. However, when the area payment is also granted to annual crops (benchmark annuity with subsidy), SRC sites solely outperform the benchmark at yield level 16 t_{atro} ha⁻¹ a⁻¹.

3.4. Cost scenarios

This section scrutinises the economic impact of low, medium and high cost levels. The respective values were derived from Table 2. The woodchip price scenario is assumed medium and subsidies are not available.

Investigating cost scenarios is very relevant as most cost items for SRC sites are likely to decrease with increasing commercial practises, i.e. the supply chain develops which results in price decreases for crop cuttings or learning effects cause decreasing plantation and harvesting cost levels. Future costs therefore largely depend on the commercial success of SRC in Germany. The low cost scenario provides reasonable estimates for such a development. It should also be noted that other cost items such as transportation expenses and fertilisers that are linked to energy prices might become more expensive (see results of high cost levels).

Fig. 4c shows the SRC annuity per cost scenarios and yield levels. When high costs are assumed, the SRC plantations perform below agricultural crops. To break even, SRC must reach yields near 12 t_{atro} ha⁻¹ a⁻¹. Under maximum yield (16 t_{atro} ha⁻¹ a⁻¹) the NPV annuity amounts to 340 €. In contrast, under a low costs scenario, NPV annuities are entirely positive, ranging from 90 € to 1000 € (6 t_{atro} ha⁻¹ a⁻¹ to 16 t_{atro} ha⁻¹ a⁻¹). The benchmark annuity can be outperformed at a yield level as low as 8 t_{atro} ha⁻¹ a⁻¹.

3.5. Comparing the economic results to other studies

The economics of SRC have already been assessed in Sweden [47], Poland [10], UK [48] and Northern Ireland [49,50]. A

comparison of findings is however challenging given differing assumptions, evaluation methods and publication dates. Nevertheless, two studies were selected as they provide economic estimates under different assumptions by applying the dynamic appraisal technique (see 2.4.2). The studies assessed the economics of SRC in Poland [10] and Northern Ireland [50]. Table 4 offers an overview on profitability under exemplary assumptions in Germany, Poland and Northern Ireland.⁹

The NPV annuity amounts to 570 € per hectare in Germany, 390 € per hectare in Northern Ireland and 260 € per hectare in Poland. This discrepancy can be mainly attributed to different woodchip prices. While German agriculturists can expect prices of 130 € per t_{atro}, average woodchip price in Northern Ireland amounts to 96 € per t_{atro} while they average 71 € per t_{atro} in Poland. Moreover, there exist significant cost differences among the countries. The cost level is the lowest in Poland with a value of 38 € per t_{atro}. In Germany and Northern Ireland, in contrast, production costs are 58 € per t_{atro} and 51 € per t_{atro} respectively.

However, it is important to remark that Ericsson et al. [10] and Dawson [50] published their papers in 2006 and 2007 respectively where woodchip prices were still at a lower level.

4. Conclusions

SRC plantations on agricultural land can contribute a significant source of wooden biomass to Europe's energy mix and therefore help Member States meet their CO₂ and renewable energy targets [12,51]. Yet, limited practical experience and uncertainties about key economic variables impede a reliable economic assessment of commercial SRC sites. This is aggravated by SRC's unfavourable cash-flow structure which is characterised by high initial investments and late amortisation. SRC plantations, relative to traditional agricultural crop, present greater investment risks, which need to be compensated by superior economics.

In order to facilitate the economic feasibility of SRC, as well as to appraise risks and opportunity costs, this study determines the profitability of SRC plantations in Germany. Under

⁹ All monetary values are in 2010 €. The relevant exchange rates were retrieved directly from the source. Lease and overheads are excluded.

the medium scenario (medium woodchip market price; absence of subsidies and medium cost level) we show that SRC plantations are profitable when site conditions are at least moderate. Notwithstanding, SRC is always outperformed by agricultural crops irrespective of the site conditions. A more detailed analysis revealed that SRC viability however increases with improved site conditions (as a result of a lower proportion of variable costs relative to that associated with traditional crops). Favourable policy instruments and economic conditions, such as high subsidies, lower costs and higher woodchip prices can lead to SRC's superior profitability (above agricultural crops).¹⁰ Finally, we show that SRC profitability is greatest in Germany if compared to Poland and Northern Ireland contexts, mainly as a result of higher woodchip prices.

In conclusion, under current investment conditions in Germany, SRC are not a viable alternative. In order to reach a SRC plantation area of 450,000 ha by 2020, as predicted in governmental scenarios analysis [12], we recommend to improve subsidy sufficiency, efficiency and consistency. Subsidies are currently insufficient: even under the most attractive subsidy scheme (Saxony's 30% establishment payment) agricultural crops economically outperform SRC. Subsidies are also inefficient in that they currently take the form of fixed payments irrespective of the produced biomass. To address this, the introduction of market-based CO₂ allowances for commercial SRC projects could contribute to alleviating this challenge and provide an additional incentive to optimise operational efforts. This could be accompanied by increased government funding for private and public SRC research with particular focus on breeding and cultivation projects to generate higher biomass levels for a given area. Finally, subsidies are inconsistent as they vary between Federal States. The legal framework for SRC is fragmented due to partly different agricultural and forestry legislations in Germany's Federal States. The alignment of regulations and guidelines is therefore crucial to improve transparency, reduce administration costs and facilitate economies of scale for operators.

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¹⁰ Based on the assumptions of high subsidies, we also conducted an exemplary Geo Information System (GIS) analysis to identify the economic potential of SRC on agricultural land in the Federal State of Saxony. The results showed that roughly 35% of the agricultural land could be planted with SRC contributing to almost 50,000 PJ of wooden biomass to Saxony's energy supply.

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